Treasury rates monitor

Task 1 | Descriptive Python analytics for a client briefing

Prepared 14 October 2026 | Retrospective report on the saved project package

Finding: the weekly endpoint understates the path

At the saved snapshot's latest complete observation, 9 October 2026, the 2-year Treasury yield was 4.80% and the 10-year yield was 5.24%. The 2s10s spread was +44 basis points. Over five complete observations, both yields fell and the spread narrowed by 1 bp: a small bull flattening.

Comparison to 9 Oct	2Y change	10Y change	Spread change
8 Oct: 1 observation	+5 bp	+2 bp	-3 bp
2 Oct: 5 observations	-3 bp	-4 bp	-1 bp
10 Sep: 21 observations	+24 bp	+29 bp	+5 bp

The weekly spread ranged from 44 to 51 bp and accumulated 13 bp of absolute close-to-close movement, despite ending only 1 bp lower. Endpoint change, range and cumulative movement answer different questions. None is portfolio profit and loss or intraday volatility.

Business question

How did outright interest rates and curve shape change, and what should a rates salesperson discuss with an institutional client? The monitor creates a reproducible factual briefing before any holdings-specific risk or hedging analysis.

The daily move was bear flattening; the longer 21-observation move was bear steepening. A horizon must therefore accompany every curve label. These are descriptions of the saved data, not explanations of why markets moved or forecasts of what comes next.

1. Data and calculation design

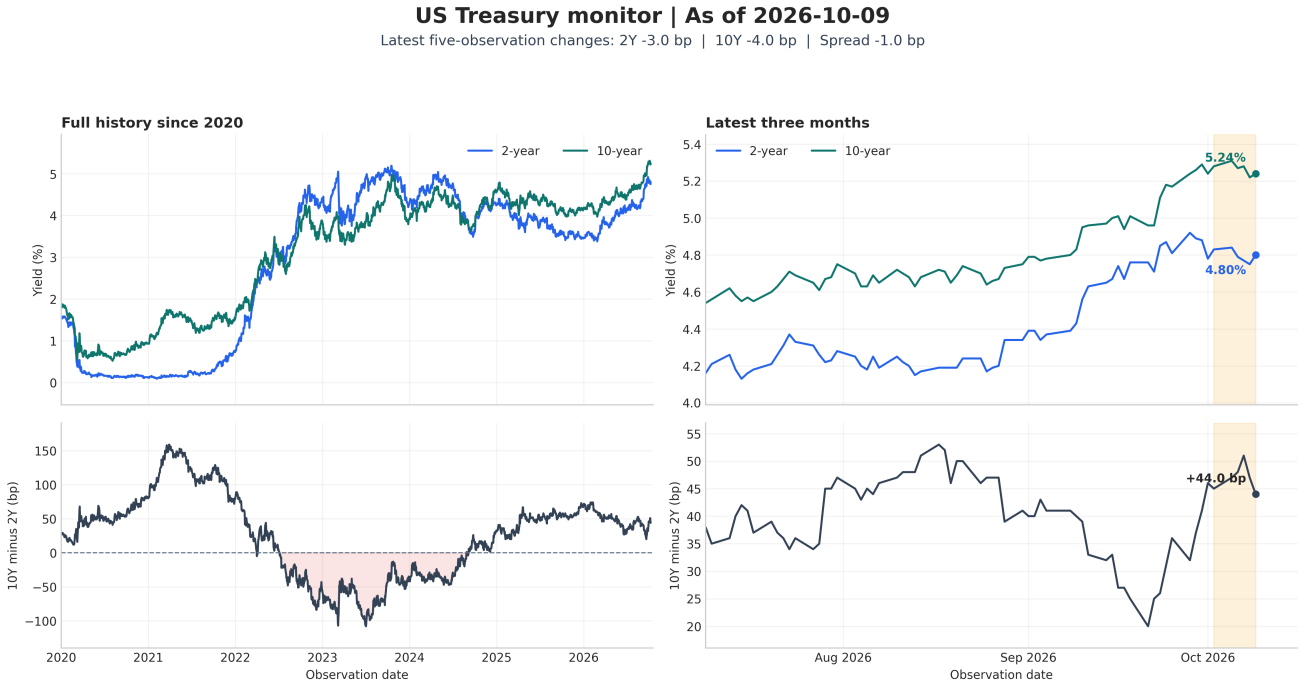
Element	Implementation
Source	Federal Reserve H.15 through FRED: DGS2 and DGS10 constant-maturity yields.
Requested window	1 January 2020 through 13 October 2026; actual complete coverage starts 2 January 2020 and ends 9 October 2026.
Units	Yield levels are percent per annum. Multiply a percentage-point difference by 100 to obtain basis points.
Spread	2s10s = 100 x (10Y yield - 2Y yield). Positive means the 10Y yield exceeds the 2Y yield.
Comparisons	1, 5 and 21 complete paired observations. These approximate daily, weekly and monthly horizons; actual dates are retained.
Missing data	Preserve missing observations for audit; exclude incomplete pairs from calculations. No forward fill.

The spread-change identity is: change in spread = change in 10Y yield - change in 2Y yield. For the weekly result, -4 - (-3) = -1 bp. The code checks this identity rather than treating spread changes as a separate data feed.

Historical percentiles compare the absolute spread move with same-horizon moves ending in the preceding two calendar years. Windows overlapping the current move are excluded and ties use midranks. Historical windows still overlap one another, so the percentile is descriptive, not an independent statistical test.

The saved weekly absolute-move percentile is 13.54, versus 69.34 for the daily move and 43.11 for the 21-observation move. These rank endpoint changes only; the low weekly percentile does not establish that the path was quiet.

2. Visual evidence and client interpretation



Read the long history for regime context and the recent window for detail. Independent vertical scales can magnify recent movement; compare axis labels before judging relative size. The chart is a historical display, not a fitted econometric model.

A client can have material outright rate risk even when the curve spread barely changes. To translate these moves into approximate portfolio value changes, request positions, maturity exposures and key-rate DV01, together with hedge details. Yield changes alone do not determine actual returns.

3. Validation, reproduction and sources

Recorded check	Saved result
Source rows / complete pairs	1,768 rows; 1,695 complete pairs; 73 rows missing both yields.
Duplicate / asymmetric missing dates	Zero duplicate dates; zero single-series missing mismatches.
Arithmetic	Spread levels and changes reconcile to underlying yields.
Reproduction	The project review records a rerun from the uploaded snapshot and visual inspection of the comparison chart.

Limits: calendar exceptions were not fully reconciled to the publication calendar. The entire numerical history was not independently checked against a second publisher. The snapshot is not a point-in-time vintage dataset and does not establish freshness after 9 October. Constant-maturity yields are not individual bond returns.

Run the original package

Extract BNY_Task1_Interview_Pack.zip. In bny_task1, install dependencies with python -m pip install pandas matplotlib, then run python task1.py --csv fred_treasury_2020-01-01_to_2026-10-13.csv --end 2026-10-13. Outputs are written to rates_monitor/outputs.

AI-assisted development: the original review records AI-assisted coding, a learner-requested expansion from daily to multiple horizons, correction of local date filtering and preservation of the raw input. This demonstrates an assisted development and verification workflow; it does not establish independently authored code or use of another AI product.

Report evidence: README.md; REVIEW_AND_PROGRESS.md; client_monitor, weekly_path_summary, methodology and briefing outputs in the saved Task 1 package. This report summarizes those artifacts; the full pipeline was not rerun for this reporting step.

Source references: https://fred.stlouisfed.org/series/DGS2 ; https://fred.stlouisfed.org/series/DGS10 . These identify the original series; the report uses the saved snapshot, not a new market-data download.